THAMIZHARASU SARAVANAN

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Executive Summary

Aspiring Artificial Intelligence and Machine Learning enthusiast with a B. Tech in Information Technology and a strong foundation in data-driven problem-solving. Experienced in implementing deep learning models with practical exposure to Python-based AI frameworks. Demonstrates a keen interest in research and innovation, backed by hands-on internships in both clinical and industrial environments. Eager to contribute technical expertise and analytical skills to forward-thinking organizations focused on AI/ML advancements.

Education

B.Tech. Information Technology (IT), *[2022 Nov - present]*
Puducherry Technological University, Puducherry, India ***CGPA (9.6** till 5th Semester / Current 7th Semester**)***

HSC, *[2021-2022]*
Sri Sankara Vidyalaya Mat. Hr. Sec. School, Puducherry, India ***93.2%***

SSLC, *[2019-2020]*
Sri Sankara Vidyalaya Mat. Hr. Sec. School, Puducherry, India ***94.6%***

Work Experiences

Project Intern, *JIPMER Puducherry* *(March 2025 - Present)*

- Developed an end-to-end pipeline for preprocessing CT scan images and training CNN/Transformer models for segmenting liver, tumour, and vascular structures.
- Utilized 3D Slicer to visualize segmented anatomical regions, enabling effective analysis and interpretation by medical professionals.

Intern, *Manatec Electronics Pvt Ltd* *(May 2024 – Jun 2024)*

- Demonstrated a high level of proficiency and commitment to the project "Parameter checking for Wheel Alignment."

Software Skills

Python *(Core Python, Flask, Streamlit)*
Data science *(NumPy, Pandas, Seaborn, Matplotlib)*
Machine Learning *(Scikit-learn, PyTorch)*
C/C++/Java Programming – *(Data structures, OOPS)*
Web Development (Basics): *(HTML, CSS, JavaScript (Basics), MySQL, React Framework (Basics), JAVA (Core JAVA, Spring Boot Framework - Basics)*

Projects

3D Liver, Tumor and Vascular Structure Segmentation, *[JIPMER, Puducherry]*

- Segmentation of Liver, Tumor, Arteries and Veins from patient's CT-SCAN images using CNNs/Transformer architectures.
- 3D reconstruction of the Segmented Liver, Tumor, Arteries and Veins for Medical Use.
- Tech stack: PyTorch, 3D UNET, D-Former, 3D Slicer, Matplotlib

End to End Solar Power Efficiency Prediction Pipeline, *[Personal Project]*

- A robust, modular, and production-ready platform for solar power system data analysis, machine learning, and prediction.
- This project enables end-to-end workflows from data ingestion and EDA to model training, deployment, and inference—all with experiment tracking and a user-friendly web interface.
- Tech stack: ZenML, ML-Flow, Streamlit

PyEdit, *[Personal Project]*

- A terminal-based AI powered text editor inspired by Vim, implemented in Python.
- Has dedicated commands like refactor, doc, explain, nl2code, translate, etc.
- This is designed for speed, efficiency, and extensibility, making it a great choice for developers who love working in the terminal.
- Tech stack: Pure Python, Curses(library), Gemini API, etc.

Courses

Joy of Computing Using Python, *NPTEL, IIT Ropar* *[Sep'2024]*
Data Science with Python, *FITA-Puducherry* *[Aug'2024]*
Introduction to Cognitive Psychology, *NPTEL, IIT Guwahati* *[Apr'2024]*
Diploma in Full Stack JAVA Development, *CSEC-Puducherry* *[Jan'2024]*
Certificate Course in Python, *CSEC-Puducherry* *[Dec'2023]*
Diploma in J2EE Programming, *CSEC-Puducherry* *[Dec'2022]*
Diploma in Computer Application, *CSEC-Puducherry* *[Sep'2019]*

Workshops & Hackathons

Workshop on Augmented Reality, *NEURA 2023- Puducherry technological University*
Workshop on Full Stack development, *ICON 2023- Puducherry technological University*

Intel Gen-AI Hackathon, *GenAI Hackathon 2024- KPR Institute of Engineering and Technology*
IndustriAI Hackathon, *SHAASTRA 2025- IIT Madras*
AWS ML Ascend Challenge – 2nd Edition, *2025 – Zelestra*

Competencies

- Problem Solving
- Team Player
- Quick Learner
- Critical Thinking